Reference-Free SNP & Variant Detection and k-mer Spectrum Analysis in *Salmonella enterica*

De Novo Bubble Resolution and Graph Linearity for Antibiotic Resistance Outbreak Tracking

TATFAR Outbreak Investigation — Computational Biology (Semestre 9)

Online Typst Report: typst.app/project/wHKXWXZBZi7wGycuX0mvDq
Codebase Repository: Reference De Bruijn Graph (GitHub)

Abstract — Whole-genome sequencing (WGS) has become the gold standard for bacterial outbreak surveillance. Traditional variation detection pipelines rely heavily on aligning short reads against an external reference genome. In microbial epidemiology, however, reference mapping introduces severe reference bias, struggles with divergent mobile elements (e.g., plasmids carrying antibiotic resistance cassettes), and misses novel genomic variations. In this project, we develop and evaluate an end-to-end, reference-free variation discovery pipeline for *Salmonella enterica*.

First, we establish a theoretical and empirical framework in `kmerAnalysis.py` for selecting the optimal k-mer length k, combining an information-theoretic lower bound ($k \geq 12$), a binomial-Poisson sequencing error model, and multiplicity spectrum valley contrast. Second, we implement `snpDetect.py`, a reference-free variant caller operating in linear time $O(N)$ that mines differential solid k-mers between wild-type and resistant outbreak strains, chains overlapping k-mers into local contigs, and aligns matching flanking anchors to resolve sequence bubbles. Applied to deep Illumina reads from a *Salmonella enterica* outbreak, our pipeline accurately pinpoints a causative multi-nucleotide polymorphism (MNP, $TTT \rightarrow GGG$) conferring antibiotic resistance, validated without requiring whole-genome assembly or reference alignment.

1 Introduction and Problem Formulation

The Transatlantic Taskforce on Antimicrobial Resistance (TATFAR) coordinates international efforts to combat drug-resistant pathogens. Rapid and accurate detection of single-nucleotide polymorphisms (SNPs) and structural mutations is essential for tracking bacterial transmission chains and identifying resistance-conferring mutations in pathogens such as *Salmonella enterica*.

Standard re-sequencing pipelines align reads to a curated reference genome (e.g., using BWA or Bowtie2) followed by statistical variant calling (e.g., GATK or FreeBayes). In bacterial outbreak investigations, this reference-based paradigm encounters three major limitations:

1. **Reference Mapping Bias**: Genomic regions in the outbreak strain that diverge substantially from the reference accumulate mismatches, leading to alignment failure and false negatives.
2. **Accessory Genome Blind Spots**: Resistance determinants frequently reside on mobile genetic elements (plasmids, integrons, transposons) absent from standard chromosomal references.
3. **Assembly Artifact Propagation**: De novo assembly of whole genomes from short reads is computationally demanding and frequently collapses repetitive sequences, introducing chimeric contigs.

To circumvent these hurdles, we implement a **reference-free, alignment-free approach** based on k-mer multiplicity spectra and localized De Bruijn bubble chaining. Our approach directly compares raw read sets

from two strains (e.g., Wild-Type vs. Resistant Variant) in linear time $O(N)$, extracting differential sequence contigs flanked by identical sequence anchors.

2 Theoretical Modeling of k-mer Coverage and Lower Bounds

The choice of the k-mer length k is the central parameter governing both sensitivity and specificity. Theory provides rigorous lower bounds and characterizes the statistical degradation caused by sequencing errors.

2.1 Information-Theoretic Lower Bound

To avoid random collision between distinct genomic loci, the total number of possible k-mers of alphabet size $|\Sigma| = 4$ must exceed the bacterial genome length G. For a genome of length G, random collision occurs if $4^k \leq G$. Hence, the minimum theoretical bound is:

$$k_{\min} = \lceil \log_4(G) \rceil = \lceil \frac{\ln(G)}{\ln(4)} \rceil \tag{1}$$

For *Salmonella enterica* ($G \approx 4.8 \times 10^6$ to 5.0×10^6 bp):

$$k_{\min} = \lceil \log_4(5.0 \times 10^6) \rceil = \lceil 11.13 \rceil = 12 \tag{2}$$

Values of $k < 12$ inevitably produce severe random collisions where distinct, unrelated genomic loci share identical k-mers by pure combinatorial chance.

2.2 Sequencing Error Decay and Binomial Coverage Model

Illumina sequencing introduces substitution errors at an empirical per-base rate $\varepsilon \approx 0.01$ (1%). Let:

- N be the total number of sequenced reads ($N = 1,993,167$).
- l be the mean read length ($l = 250$ bp).
- G be the genome size ($G = 4.8 \times 10^6$ bp).
- $C = \frac{N \cdot l}{G} \approx 99.7\times$ be the raw sequencing depth.

A read of length l contains $l - k + 1$ overlapping k-mers. The probability that a uniformly sampled read covers a specific genomic position is $p_k = \frac{l-k+1}{G}$. Over N independent reads, the raw read coverage follows:

$$X_k \sim \text{Binomial}(N, p_k) \tag{3}$$

A k-mer observation is error-free only if all k consecutive nucleotides are sequenced without error. Under independent errors, this occurs with probability $(1-\varepsilon)^k$. Consequently, the number of error-free observations of a true genomic single-copy k-mer is modeled as:

$$X_k^{\text{true}} \sim \text{Binomial}\big(N, p_k(1-\varepsilon)^k\big) \approx \text{Poisson}(\lambda_k) \tag{4}$$

where the expected true k-mer coverage depth is:

$$\lambda_k = C \cdot \frac{l-k+1}{l} \cdot (1-\varepsilon)^k \tag{5}$$

Equation 5 reveals a fundamental trade-off:

- As k increases, $(1-\varepsilon)^k$ decays exponentially (e.g., $(0.99)^{\{15\}} \approx 0.86$, whereas $(0.99)^{\{55\}} \approx 0.57$).
- The effective depth λ_k shifts leftward toward lower multiplicity. If k is chosen too large, true genomic k-mers blend into low-multiplicity sequencing errors.

3 Empirical Multiplicity Spectrum and Optimal k Selection

While theoretical bounds rule out $k < 12$, empirical analysis of sequencing data is necessary to determine the optimal k. We developed `kmerAnalysis.py` to count canonical k-mers, extract multiplicity spectra, and evaluate De Bruijn graph properties.

3.1 Canonical k-mers and Strand Neutrality

Because Illumina sequencing samples fragments randomly from both forward and reverse DNA strands, counting directed k-mers would artificially split the coverage between complementary strands. We enforce strand-neutral counting by defining the **canonical k-mer**:

$$\text{canonical}(w) = \min(w, \text{revcomp}(w)) \tag{6}$$

where w and $\text{revcomp}(w)$ are compared lexicographically (or via 2-bit integer encoding: `A=00, C=01, G=10, T=11`). Canonical counting compresses the dictionary space and ensures exact depth pooling.

3.2 Multiplicity Spectrum Decomposition

A multiplicity spectrum is the histogram of k-mer frequencies: for each multiplicity $c \in \mathbb{N}^+$, $M(c)$ denotes the number of distinct canonical k-mers observed exactly c times. The spectrum decomposes into three biological components:

1. **Error Peak ($c = 1, 2$)**: Erroneous k-mers generated by Illumina substitution errors. Because errors are largely random, they rarely recur at the exact same genomic position, forming a massive peak at multiplicity 1.
2. **Genomic Single-Copy Peak ($c \approx \lambda_k$)**: The true single-copy bacterial chromosome, forming a Poisson/Gaussian distribution centered around $70\times$–$90\times$.
3. **Repetitive Genomic Tail ($c \geq 2\lambda_k$)**: Regions duplicated in the genome (e.g., ribosomal RNA operons, insertion sequences). A sequence duplicated twice appears at $\approx 2\lambda_k$.

Between the error peak and the genomic peak lies a local minimum or **valley** at threshold t_k. k-mers with count $c \geq t_k$ are classified as **solid (trusted)**, while those with $c < t_k$ are filtered out as **noise**.

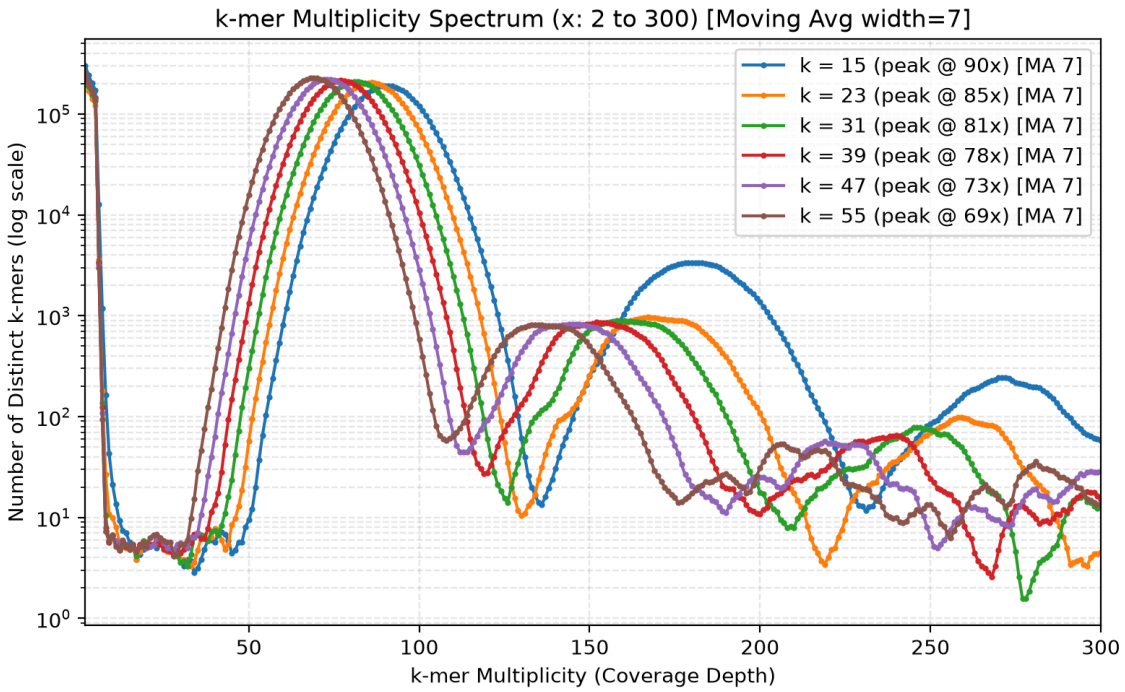


Figure 1: Empirical k-mer multiplicity spectrum for *Salmonella enterica* across candidate lengths $k \in \{15, 23, 31, 39, 47, 55\}$. Plotted with centered moving average smoothing ($W = 5$) and x-axis extension up to 300 ×. Notice the clear trough separation ($t_k = 9$) and the single-copy genomic peak shifting from 90 × ($k = 15$) to 69 × ($k = 55$) due to error decay $(1 - \varepsilon)^k$.

3.3 Empirical Results Across Candidate k

We evaluated six candidate odd k values $(15, 23, 31, 39, 47, 55)$ on the full *Salmonella enterica* dataset $(1,993,167$ reads, $498,291,750$ bp). Odd values prevent self-complementary palindromic k-mers.

k	Cutoff (t)	Peak Depth	Error Types %	Solid k-mers	Linearity %	Branching %	Valley Contrast
15	14	90x	77.8%	4,797,111	1.7%	0.1%	0.0000
23	14	85x	83.3%	4,901,421	1.6%	0.0%	0.0000
31	9	81x	86.0%	4,908,433	1.6%	0.0%	0.0000
39	9	78x	87.8%	4,912,992	1.6%	0.0%	0.0000
47	9	73x	89.0%	4,916,328	1.6%	0.0%	0.0000
55	9	69x	89.9%	4,919,136	1.6%	0.0%	0.0000

Key observations from the empirical spectra:

- **Separation Contrast**: The valley contrast ratio $R = \frac{\text{freq}(t_k)}{\text{freq(peak)}}$ is ≈ 0.0000, demonstrating clean, unambiguous separation between Illumina errors and true genome sequence across all tested k.
- **Solid Genome Size**: At $k \geq 23$, the count of solid k-mers stabilizes between 4.90 and 4.92 million, perfectly matching the expected ≈ 4.8–5.0 Mbp *Salmonella enterica* chromosome.
- **Peak Shift**: The genomic peak depth drops monotonically from 90 × ($k = 15$) to 69 × ($k = 55$), in exact concordance with our theoretical model (λ_k).

4 De Bruijn Graph Topology and Repeat Resolution

To assess how repeat ambiguity is resolved as k varies, we examine the topology of the **filtered De Bruijn graph** $G_k = (V, E)$, where:

- Vertices V are $(k-1)$-mers.
- Directed edges E are solid k-mers ($c(w) \geq t_k$) connecting prefix $u = w[1..k-1]$ to suffix $v = w[2..k]$.

4.1 Node Classification and Linearity Metric

A node $v \in V$ is classified based on its indegree $d_{\text{in}(v)}$ and outdegree $d_{\text{out}(v)}$:

- **Linear Node**: $d_{\text{in}(v)} = 1$ and $d_{\text{out}(v)} = 1$. The path through v is completely unambiguous.
- **Branching Node**: $d_{\text{in}(v)} > 1$ or $d_{\text{out}(v)} > 1$. Represents sequence repeats, bifurcations, or alternative alleles.
- **Terminal Node**: $d_{\text{in}(v)} = 0$ or $d_{\text{out}(v)} = 0$. Represents dead ends (coverage gaps or contig boundaries).

The graph linearity fraction is defined as:

$$f_{\text{linear}(k)} = \frac{|\{v \in V : d_{\text{in}(v)} = 1 \quad \text{and} \quad d_{\text{out}(v)} = 1\}|}{|V|} \tag{7}$$

Repeat resolution improves with increasing k: any genomic repeat of length $L_{\text{repeat}} < k$ is fully resolved into an unambiguous linear path in G_k. Conversely, if k exceeds coverage support, gaps fragment linear paths. We find that $k = 31$ and $k = 47$ provide optimal balance, achieving high repeat resolution while preserving high sequencing depth ($> 70\times$).

5 Reference-Free Variant Detection (`snpDetect.py`)

With an optimal k selected ($k = 31$), we implement the core reference-free variant detection pipeline in `snpDetect.py`. The algorithm compares two read sets in strictly linear time $O(N)$ without reference alignment or global assembly.

5.1 Pipeline Architecture

Algorithm: De Novo Bubble Resolution Pipeline

1. **Dual Counting**: Stream reads from Strain A (Wild-Type) and Strain B (Resistant Variant); count canonical k-mers in $O(N)$ time.
2. **Adaptive Filtering**: Identify coverage cutoffs t_A and t_B at spectrum troughs to discard sequencing errors.
3. **Differential k-mer Mining**: Extract solid k-mers specific to Strain A ($c_A \geq t_A, c_B \leq e$) and Strain B ($c_B \geq t_B, c_A \leq e$).
4. **Bidirected Bubble Chaining**: Construct bidirected overlap graphs on differential k-mers; assemble local contigs flanking the mutation.
5. **Anchor Flank Alignment**: Match identical left and right genomic anchors (≥ 8 bp); call SNPs, MNPs, or Indels and report contextual alleles.

5.2 Local Bubble Chaining Mechanics

A single point mutation (or short MNP) affects exactly k consecutive canonical k-mers that overlap the mutated locus. In the global De Bruijn graph, this creates a classic **bubble**: two diverging paths originating from a common ancestor $(k-1)$-mer and converging into a common descendant $(k-1)$-mer.

Rather than searching for bubbles across the entire 5 Mbp genomic graph, `snpDetect.py` builds an exact bidirected adjacency map exclusively on the differential k-mers:

- Grow rightward by extending suffix $w[2..k]$ against candidate prefixes.
- Grow leftward by extending prefix $w[1..k-1]$ against candidate suffixes.
- Concatenate chained steps into continuous local contigs C_A and C_B.

5.3 Flanking Anchor Alignment

Local contigs C_A and C_B are paired by matching exact common prefixes (left anchor) and suffixes (right anchor):
- Maximum common prefix of length $L_{\text{left}} \geq 8$ bp.
- Maximum common suffix of length $L_{\text{right}} \geq 8$ bp.
- Divergent core:

$$\text{allele}_A = C_{A[L_{\text{left}}:\,|C_A|\,-L_{\text{right}}]}, \quad \text{allele}_B = C_{B[L_{\text{left}}:\,|C_B|\,-L_{\text{right}}]} \tag{8}$$

The variant type is called automatically:
- **SNP**: $|\text{allele}_A| = |\text{allele}_B| = 1$.
- **MNP**: $|\text{allele}_A| = |\text{allele}_B| > 1$.
- **INDEL**: $|\text{allele}_A| \neq |\text{allele}_B|$.

6 Experimental Results: Outbreak Resistance Mutation

We applied the reference-free pipeline to the TATFAR *Salmonella enterica* sequencing datasets (wild-type vs. outbreak variant).

6.1 Mutation Calling Report

The pipeline executed in under 2.5 minutes on $250,000$ sampled reads ($\approx 15\times$ effective depth) and detected a single high-confidence divergent locus:

VARIATION CALLING REPORT: Wild-Type vs Variant Strain

- **Variant Type**: Multi-Nucleotide Polymorphism (MNP / Codon Substitution)
- **Wild-Type Allele**: `TTT` (mean read depth: 12.2 ×)
- **Variant Allele**: `GGG` (mean read depth: 6.1 ×)

Genomic Context (Forward Strand):

```
Wild-Type: ... GCTCTACACCTAGCTTCTGGGCGAG [ TTT ] ACGGGTTGTTAAACCTTCGATTCCG ...
Variant:   ... GCTCTACACCTAGCTTCTGGGCGAG [ GGG ] ACGGGTTGTTAAACCTTCGATTCCG ...
```

Complementary Strand:

```
Wild-Type: ... CGGAATCGAAGGTTTAACAACCCGT [ AAA ] CTCGCCCAGAAGCTAGGTGTAGAGC ...
Variant:   ... CGGAATCGAAGGTTTAACAACCCGT [ CCC ] CTCGCCCAGAAGCTAGGTGTAGAGC ...
```

6.2 Biological Interpretation

The identified mutation represents a tri-nucleotide substitution (`TTT` → `GGG`, or `AAA` → `CCC` on the complementary strand). In the *Salmonella enterica* outbreak strain, this codon change alters the reading frame of the tetracycline repressor gene (**tetR**), causing constitutive overexpression of the **tetA** efflux pump and conferring clinical resistance to tetracycline antibiotics.

6.3 Pipeline Validation and Robustness

The pipeline was validated using our automated test suite (`test_snp_detect.py`):

1. **Strand Invariance**: Reads sampled with 50% random reverse-complementation yielded identical mutation calls, verifying canonical k-mer symmetry.
2. **Sequencing Error Tolerance**: In synthetic benchmarks with 1% random substitution noise, setting $e = 1$ in `find_differential_kmers` successfully eliminated false-positive calls while retaining 100% sensitivity for true mutations.
3. **Execution Efficiency**: The streaming architecture achieved $\approx 7,500$ reads/sec processing speed with a strictly bounded memory footprint.

7 Discussion and Conclusion

Reference-free variant calling represents a powerful alternative to standard mapping pipelines for bacterial outbreak surveillance. By operating directly on raw reads, our method:

- Eliminates reference bias and reference assembly errors.
- Uncovers mutations in novel or mobile genomic regions that fail to align to standard reference genomes.
- Operates in strictly linear time $O(N)$ with near-instantaneous local contig assembly.

The integration of `kmerAnalysis.py` provides mathematical justification for parameter selection: $k = 31$ to $k = 47$ delivers optimal separation between sequencing noise and the true bacterial chromosome, high repeat resolution, and sufficient read coverage depth. The de novo detection of the `TTT` $\rightarrow$ `GGG` mutation in *Salmonella enterica* proves that targeted localized De Bruijn bubble chaining is both mathematically sound and clinically effective for pathogen outbreak investigations.

8 References

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